

It remains to prove the lemma.  
Proof (Lemma). We know that

$$\text{Col}(Y) \leq \frac{1}{|Y|} \cdot (1 + 4\varepsilon^2)$$

$$\text{SD}(Y; U) = \frac{1}{2} \sum_{y \in Y} \left| P_Y[Y=y] - \frac{1}{|Y|} \right|$$

Let us call  $q_y = P_Y[Y=y] - 1/|Y|$

$$s_y = \begin{cases} +1 & \text{if } q_y \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

$$\text{SD}(Y; U) = \frac{1}{2} \sum_{y \in Y} q_y s_y$$

$$\vec{q} = (q_y)_{y \in Y} \quad ; \quad \vec{s} = (s_y)_{y \in Y}$$

$$= \frac{1}{2} \langle \vec{q}, \vec{s} \rangle$$

$$\leq \frac{1}{2} \sqrt{\langle \vec{q}, \vec{q} \rangle \langle \vec{s}, \vec{s} \rangle}$$

CAUCHY  
SCHWARTZ

$$\sum_{y \in Y} s_y^2$$

$$= \frac{1}{2} \sqrt{\sum_{y \in Y} q_y^2 |Y|}$$

Let's focus on  $\sum_{y \in Y} q_y^2$ :

$$\sum_{y \in Y} q_y^2 = \sum_{y \in Y} \left( \Pr[Y=y] - \frac{1}{|Y|} \right)^2$$

$$= \sum_{y \in Y} \left( \Pr[Y=y]^2 + \frac{1}{|Y|^2} - 2 \frac{\Pr[Y=y]}{|Y|} \right)$$

$$= \text{Col}(Y) + \frac{1}{|Y|} - \frac{2}{|Y|}$$

$$\leq \frac{(1 + 4\varepsilon^2)}{|Y|} - \frac{1}{|Y|} = \frac{4\varepsilon^2}{|Y|}$$

$$\Rightarrow SD(Y; V) \leq \frac{1}{2} \sqrt{\frac{4\varepsilon^2}{|Y|}} |Y|$$

$$= \varepsilon$$

□

## COMPUTATIONAL SECURITY

As we have seen, until now there barriers to deploy provably secure and efficient GPPs. We need to change smt. We will further assume:

(i) The Adv is on "efficient" machine.

(ii) There are problems that seem "hard" i.e. no "efficient" machine can solve them!

Let's expand on those assumptions.

(i) Adv  $\equiv$  Turing Machine. Not only that, but also an "efficient" one. In particular bounded running time: POLYNOMIAL TIME!

We measure # of steps as a function of input length:  $O(n^c)$  for some  $c \in \mathbb{N}$ .

Additionally: Adv can generate arbitrary (polynomial) amount of RANDOMNESS.

$\Rightarrow$  Adv  $\equiv$   $\begin{matrix} P & P^T \\ T & M \end{matrix}$  (Probabilistic polynomial Time)

(22) Ideally,  $P \neq NP$ .

$P \equiv$  All problems solvable by PT TMs.

$NP \equiv$  All problems s.t. a solution can be verified by a PT TM.

(e.g. FACTORING :  $n = p \cdot q$   
3 COL :  $G$ ,  $\varphi$  a 3-coloring of  $G$ )

Why this would be cool? See we

can prove:

THM If  $P \neq NP$ , then " $\Pi$ " is "secure".

Only 2 options : 1)  $\Pi$  is secure

2)  $P = NP$ .

Bad news : Nobody knows how to do this.

What we know, e.g.

THM If FACTORING is hard, then  $\Pi$  secure.

Still cool? Well, if  $\Pi$  not secure then

FACTORING not hard  $\Rightarrow \exists PPT$  TM solving FACTORING.

The mother of all These assumptions:

ONE-WAY FUNCTIONS  
(OWF)

OWF are the minimal Assumption to do interesting things (as we'll see).  
deterministic

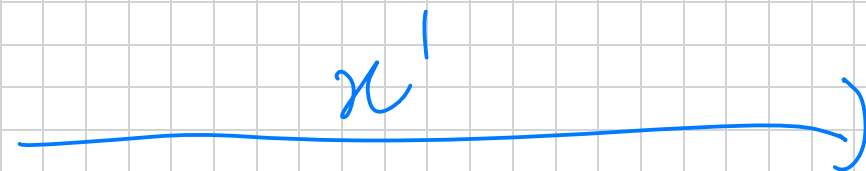
DEF A function  $f: \mathcal{X} \rightarrow \mathcal{Y}$  is

one-way if  $\forall$  PPT  $A$ :

$$\Pr [G_{\text{OWF}}^{A, f}(\lambda) = 1] \text{ is small}$$
$$\Pr [w_{\text{in}}] \leq \text{negl}(\lambda)$$

GAME<sup>out</sup>  
A, f (1)

A



$\mathcal{X}$

$$x \leftarrow \mathcal{X}$$

$$y = f(x)$$

Output 1  
(WIN)

$$f(x') = y$$

$$f: \{0,1\}^n \rightarrow \{0,1\}^n$$

How to define small?? Asymptotics!

A function  $\varepsilon(n)$  is negligible in  $n$

if  $\varepsilon(\lambda) = O(1/p(\lambda))$  for every  
poly normal  $p(\lambda) = \text{poly}(\lambda)$ .

EXAMPLES :  $z^{-1} = \text{negl}(\lambda)$ .

Why not:  $\text{negl}(\lambda) + \text{negl}(\lambda) = \text{negl}(\lambda)$ .

$\text{poly}(\lambda) \cdot \text{negl}(\lambda) = \text{negl}(\lambda)$ .

Examples of OWFs : FACTORING, DISCRETE

LOGS, LATTICES, DECODING RANDOM  
CODES

To conclude : There are different computer network  
works

